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Exercise 6 (21 points) – individual work

- The answers can be typed or handwritten (handwriting must be clear and readable), in this exercise sheet or your own sheet (put your name & ID at the top of the sheet). All ans0077ers must be saved to only 1 PDF file.
- Some questions also require the submission of processes/workflows (file.rmp or file.ipynb).
- In case of re-submission (after first grading) or submission after solution is given, your points will be weighted by 0.5.

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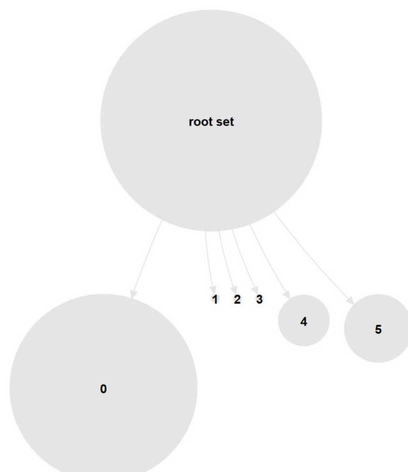
1. (Total 7 points) Retrieve **wholesale** data. Each record represents annual purchases of a wholesaler's customer.

Attribute	Short description
ID	Customer ID
Fresh	Annual purchase of fresh products
Milk	Annual purchase of milk products
Grocery	Annual purchase of groceries
Frozen	Annual purchase of frozen food
Detergents_Paper	Annual purchase of detergent & paper products
Delicatessen	Annual purchase of ready-to-eat foods

1.1 (5 points) Create workflow for K-Means clustering. Submit it with the name **question1.rmp**.

Instructions	Answer
Step 1. Apply necessary preprocessing e.g. normalization. Then, run K-Means using K = 2-10. You can do it manually or use "Optimize Parameters (Grid)" operator.	

<table>
<tr>
<td>Step 2. Report/capture DBIs for different K values. And based on these DBIs, which K is chosen?</td>
<td>Show DBIs for K = 2-10
<table>
<tr><th>iteration ↑</th><th>Cluster...</th><th>Davies...</th></tr>
<tr><td>1</td><td>2</td><td>-1.182</td></tr>
<tr><td>2</td><td>3</td><td>-0.961</td></tr>
<tr><td>3</td><td>4</td><td>-1.143</td></tr>
<tr><td>4</td><td>5</td><td>-0.904</td></tr>
<tr><td>5</td><td>6</td><td>-0.824</td></tr>
<tr><td>6</td><td>7</td><td>-0.921</td></tr>
<tr><td>7</td><td>8</td><td>-0.973</td></tr>
<tr><td>8</td><td>9</td><td>-0.941</td></tr>
<tr><td>9</td><td>10</td><td>-0.826</td></tr>
</table>
Chosen K = 6
because 6 is neareast to 0 → better seperation</td>
</tr>
<tr>
<td>Step 3. Base on the chosen K, your workflow must output cluster model (i.e. centroid plots/tables).</td>
<td>
<table>
<tr><th>Attribute</th><th>cluster_0</th><th>cluster_1</th><th>cluster_2</th><th>cluster_3</th><th>cluster_4</th><th>cluster_5</th></tr>
<tr><td>Fresh</td><td>-0.226</td><td>0.792</td><td>0.388</td><td>-0.532</td><td>1.965</td><td>1.903</td></tr>
<tr><td>Milk</td><td>-0.376</td><td>0.561</td><td>3.939</td><td>0.635</td><td>5.170</td><td>-0.094</td></tr>
<tr><td>Grocery</td><td>-0.439</td><td>-0.011</td><td>4.017</td><td>0.913</td><td>1.286</td><td>-0.178</td></tr>
<tr><td>Frozen</td><td>-0.097</td><td>9.242</td><td>0.036</td><td>-0.351</td><td>6.893</td><td>0.618</td></tr>
<tr><td>Detergents_Paper</td><td>-0.401</td><td>-0.464</td><td>4.221</td><td>0.924</td><td>-0.554</td><td>-0.380</td></tr>
<tr><td>Delicatessen</td><td>-0.189</td><td>0.932</td><td>0.941</td><td>0.026</td><td>16.460</td><td>0.389</td></tr>
</table>
</td>
</tr>
<tr>
<td>Step 4. Base on the chosen K, your workflow must output cluster profiles in the form of rules/trees or summary statistics (at least attribute means) of the <u>original attributes before preprocessing</u>.</td>
<td>
</td>
</tr>
</table>

1.2 (2 points) From the results in Steps 3 & 4 of question 1.1, choose 1 cluster. Briefly discuss its pattern, e.g. how this cluster differs from the others, which attributes (identify 1-2 attributes) set it apart from the others.

Cluster ID (from centroid plot) = cluster_4
Characterizing attributes = Delicatessen Discussion = This attribute's value rises sharply from around 0 to 16.5, the highest in the

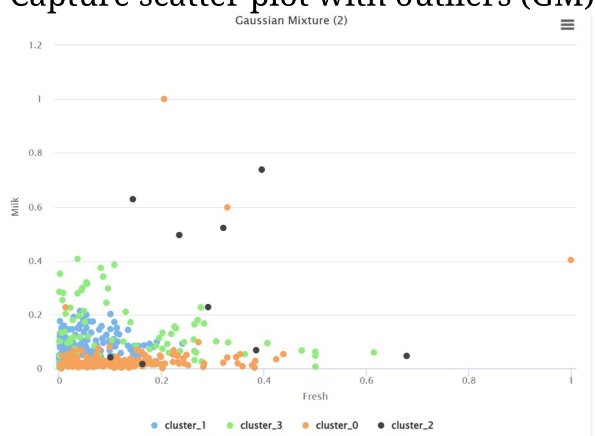
chart. Most clusters stay fairly steady, but cluster 4 has a big jump in this attribute, showing it as an extreme outlier.

2. (Total 8 points) Use the following methods on **wholesale** data in question 1.

2.1 (6 points) Run Gaussian Mixture (GM).

<table>
<tr><th>Instructions</th><th>Answer</th></tr>
<tr><td>Step 1. Run GM with BIC optimization and inverted likelihood scoring.

Note: you may get error when choosing GM's normalization parameter. In that case, apply 0-1 normalization separately before the GM.</td><td>Number of clusters discovered by GM = 4</td></tr>
<tr><td>Step 2. Set an outlier threshold such that records with very high inverted likelihood scores are deemed outliers.</td><td>Show outlier records with their IDs and scores (GM)
<table>
<tr><th>Row No.</th><th>id</th><th>confidence...</th><th>confidence...</th><th>confidence...</th><th>confidence...</th><th>cluster</th><th>score ↓</th></tr>
<tr><td>184</td><td>184</td><td>0</td><td>0</td><td>1</td><td>0</td><td>cluster_0</td><td>124.699</td></tr>
<tr><td>87</td><td>87</td><td>0</td><td>0</td><td>1</td><td>0</td><td>cluster_0</td><td>67.402</td></tr>
<tr><td>334</td><td>334</td><td>0</td><td>0</td><td>1.000</td><td>0.000</td><td>cluster_0</td><td>66.048</td></tr>
<tr><td>182</td><td>182</td><td>0</td><td>0</td><td>1.000</td><td>0.000</td><td>cluster_0</td><td>62.136</td></tr>
<tr><td>326</td><td>326</td><td>0</td><td>0</td><td>1</td><td>0</td><td>cluster_2</td><td>32.854</td></tr>
<tr><td>86</td><td>86</td><td>0</td><td>0</td><td>1.000</td><td>0.000</td><td>cluster_2</td><td>14.923</td></tr>
<tr><td>110</td><td>110</td><td>0</td><td>0.000</td><td>0.000</td><td>1.000</td><td>cluster_0</td><td>6.793</td></tr>
<tr><td>72</td><td>72</td><td>0</td><td>0</td><td>1.000</td><td>0.000</td><td>cluster_2</td><td>6.360</td></tr>
<tr><td>94</td><td>94</td><td>0.000</td><td>0</td><td>1.000</td><td>0.000</td><td>cluster_2</td><td>5.741</td></tr>
<tr><td>24</td><td>24</td><td>0</td><td>0</td><td>1.000</td><td>0.000</td><td>cluster_2</td><td>3.744</td></tr>
<tr><td>88</td><td>88</td><td>0</td><td>0</td><td>1.000</td><td>0.000</td><td>cluster_2</td><td>3.146</td></tr>
<tr><td>48</td><td>48</td><td>0</td><td>0</td><td>1.000</td><td>0.000</td><td>cluster_2</td><td>1.449</td></tr>
<tr><td>126</td><td>126</td><td>0.000</td><td>0</td><td>0.137</td><td>0.863</td><td>cluster_2</td><td>1.351</td></tr>
<tr><td>339</td><td>339</td><td>0.003</td><td>0</td><td>0.044</td><td>0.953</td><td>cluster_3</td><td>1.237</td></tr>
</table></td></tr>
<tr><td>Step 3. Locate the outliers in Step 2 in at least 1 scatter plot where they are clearly separated from inliers.

Note: check from scatter matrix first to see which pair of attributes gives clear patterns of outliers.</td><td>Capture scatter plot with outliers (GM)

</td></tr>
</table>

<table>
<tr>
<td>Step 4: Use either distance-based or LOF to detect outliers instead of GM.</td>
<td>Chosen method = distance-based

Show outlier records with their ID and scores
(other method)
<table>
<tr><th>Row No.</th><th>id</th><th>outlier ↓</th><th>Fresh</th><th>Milk</th><th>Grocery</th><th>Frozen</th><th>Detergents...</th><th>Delicatessen</th></tr>
<tr><td>24</td><td>24</td><td>true</td><td>0.235</td><td>0.495</td><td>0.237</td><td>0.084</td><td>0.106</td><td>0.345</td></tr>
<tr><td>48</td><td>48</td><td>true</td><td>0.396</td><td>0.738</td><td>0.599</td><td>0.127</td><td>0.592</td><td>0.135</td></tr>
<tr><td>62</td><td>62</td><td>true</td><td>0.320</td><td>0.522</td><td>0.642</td><td>0.053</td><td>0.654</td><td>0.042</td></tr>
<tr><td>86</td><td>86</td><td>true</td><td>0.144</td><td>0.628</td><td>1</td><td>0.016</td><td>1</td><td>0.061</td></tr>
<tr><td>87</td><td>87</td><td>true</td><td>0.204</td><td>1</td><td>0.346</td><td>0.016</td><td>0.492</td><td>0.019</td></tr>
<tr><td>94</td><td>94</td><td>true</td><td>0.101</td><td>0.041</td><td>0.022</td><td>0.575</td><td>0.002</td><td>0.056</td></tr>
<tr><td>182</td><td>182</td><td>true</td><td>1</td><td>0.403</td><td>0.196</td><td>0.275</td><td>0.121</td><td>0.178</td></tr>
<tr><td>184</td><td>184</td><td>true</td><td>0.329</td><td>0.598</td><td>0.217</td><td>0.600</td><td>0.006</td><td>1</td></tr>
<tr><td>326</td><td>326</td><td>true</td><td>0.292</td><td>0.228</td><td>0.147</td><td>1</td><td>0.031</td><td>0.117</td></tr>
<tr><td>334</td><td>334</td><td>true</td><td>0.076</td><td>0.067</td><td>0.725</td><td>0.002</td><td>0.933</td><td>0.025</td></tr>
</table>
</td>
</tr>
<tr>
<td>Also submit the workflow that performs Steps 1-4. Name your workflow question2.rmp.</td>
<td></td>
</tr>
</table>

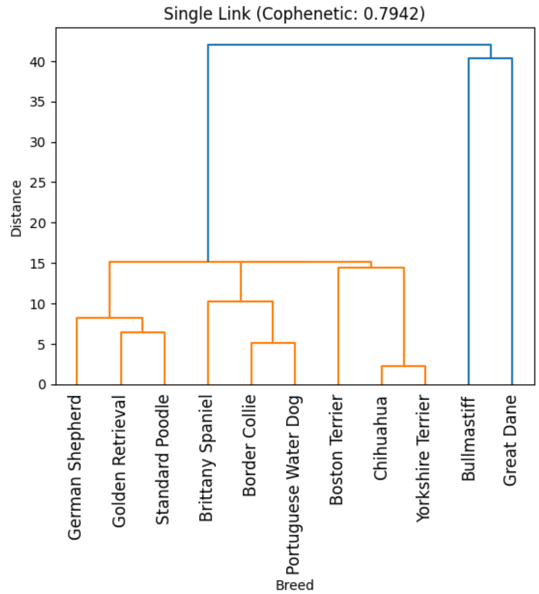
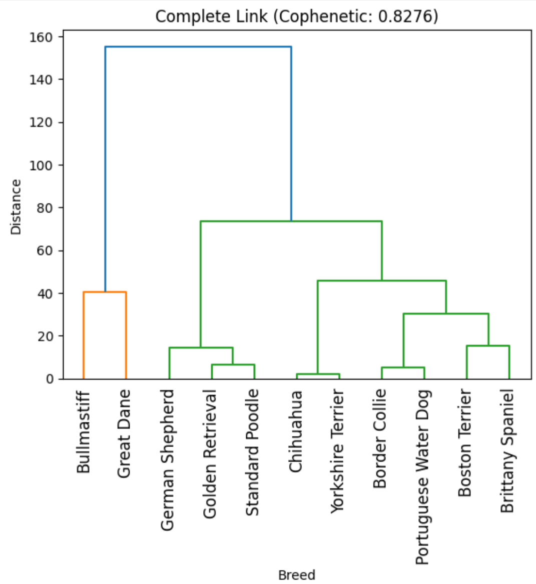
2.2 (2 points) Pick 2 records that you suspect them to be outliers. They can be outliers found by any method in question 2.1 or even from clustering result in question 1.1. Inspect their attribute values. Report which of their values are very high or very low compared to the values of inliers.

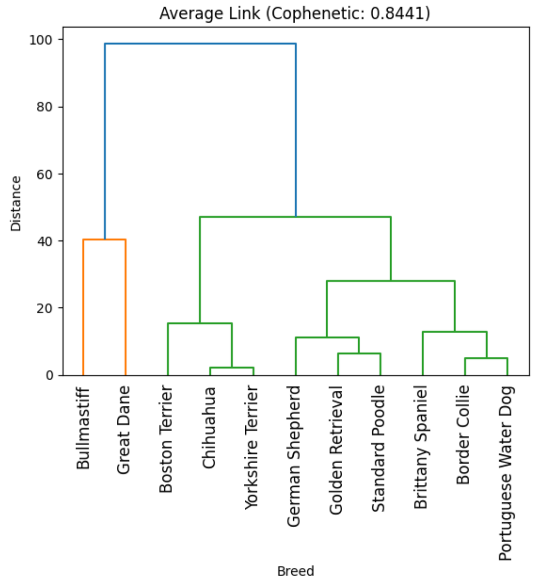
Record ID	Observations on attribute values
Outlier 1 = Record 182	The fresh value is very high compared to other values which is in range.
Outlier 2 = Record 86	The Grocery and Detergents_Paper values are very high compared to other values which is in range.

3. (Total 6 points) Use python functions to run hierarchical clustering (HAC) on **dogbreeds.csv**. Compare the following types of linkage.

3.1 (3 points) Capture dendrogram & report cophenetic correlation of each linkage type.

Linkage	Dendrogram & cophenetic correlation

Single link	
Complete link	

Group average (i.e. average link)	

3.2 (1 point) Which linkage type gives clusters that best fit the data?

Group average gives clusters the best fit.

3.3 (2 points) Consider result from the best linkage type in (3.2), if we cut the dendrogram into 2 clusters, which dog breeds will belong to which clusters.

Cluster	Dog breeds
Cluster 1	Bullmastiff, Great Dane
Cluster 2	Border Collie, Boston Terrier, Brittany Spaniel, Chihuahua, German Shepherd, Golden Retrieval, Portuguese Water Dog, Standard Poodle, Yorkshire Terrier